AVIKAM JHA

Design Verification Engineer
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SUMMARY

Computer Engineering M.S. candidate (**GPA 4.0**, Dec 2026) focused on **design verification**. Built a **UVM testbench** for a pipelined LC-3 processor that reached **99.13% coverage**, co-wrote **verification plans** from a specification, built SystemVerilog testbenches, closed **functional coverage** with cross coverage, and checked the DUT against an independent **C reference model**. Brings CPU micro-architecture depth from designing and debugging a pipelined CPU, gate-level verification from an RTL-to-GDS accelerator, and **2+ years as a software engineer** with Python, Shell, and Linux.

EDUCATION

Arizona State University, Tempe, AZ | M.S., Computer Engineering | GPA 4.0/4.0 | Dec 2026
Relevant coursework: VLSI Design, Digital Systems and Circuits, SoC Design, Computer Architecture II, Digital Verification and Test

National Institute of Technology, Delhi | B.E., Electronics and Communication Engineering | July 2022

TECHNICAL SKILLS

Verification & Simulation Tools: SystemVerilog, **UVM** (Siemens UVMF), Siemens Questa and Visualizer, UCDB coverage merging, ModelSim, Xilinx Vivado (simulation), PLI/C co-simulation; SVA and SymbiYosys formal in progress

Design & Architecture Tools: Verilog, Synopsys Design Compiler, Cadence Innovus, gem5, CUDA

Scripting & Languages: Python, C, C++, Shell/Bash, Make, Tcl

Platforms: Linux, Git, Kubernetes

PROJECTS

LC-3 Pipelined Processor Verification with UVM | SystemVerilog, UVM (UVMF), Questa | Spring 2025

- Built a UVM testbench for a **5-stage pipelined LC-3** processor with a dedicated **environment per pipeline stage** (Fetch, Decode, Execute, MemAccess, Writeback), each with sequencer, driver, monitor, **predictor**, and **scoreboard** across **14 interface agents**.
- Wrote reference-model predictors for every stage and **covergroups with crosses** over ALU, memory (LD, LDI, LEA, LDR, ST, STI, STR), and control (BR, JMP) instructions, covering all valid registers, offsets, and NZP combinations.
- Reached **99.13% coverage** and met every test-plan goal with **12 seeded tests** in a scripted Questa regression that merges the XML test plan and simulation results into one UCDB.

VeriChip: Verification Plans, Testbenches & Functional Coverage | SystemVerilog, C | 2025

- Reviewed a memory-mapped peripheral specification and **co-authored test and coverage plans** for the register, state-machine, and ALU blocks, tracing every planned test to a section of the spec.
- Exhaustively verified a **4-state FSM** across every input combination plus **error conditions** (overflow, bad command, export violation) in every state, and all **16 ALU commands** across export-enabled and export-disabled modes.
- Closed coverage with **four covergroups** sampling on both clock edges, state-conditional coverpoints, and **3-way crosses** across command, chip-select, read/write, byte enables, and seven register addresses.
- Wrote a **C reference model** linked through PLI that reads live DUT outputs and computes expected values independently of the RTL, with seed-controlled random stimulus for repeatable failure reproduction.

5-Stage Pipelined 32-bit CPU | SystemVerilog | Aug – Dec 2025

- Built a 32-bit RISC-style core from memory, register file, ALU, and PC primitives to a full pipeline with **ID/EX, EX/MEM, and MEM/WB** registers and a hazard-detection control FSM.
- Designed a **forwarding unit** with EX/MEM-over-MEM/WB priority and three independent bypass networks (ALU operands, branch comparands, jump-register targets), resolving **five stall conditions** across load-use, jump-register, and branch hazards.
- Extended the ISA with **load-linked/store-conditional** atomics and built a **CAM** for single-cycle cache tag lookup.

Cross-Stage Verification of an AI Accelerator: RTL to Post-Layout Netlist | SystemVerilog, Design Compiler, Innovus, ASAP7 7nm | 2025

- Re-ran one testbench at **RTL, post-synthesis, and post-APR** netlist stages on a 16,590-cell sparse GCN accelerator to separate timing-induced mismatches from functional bugs.
- Verified behavior through micro-architecture changes (zero-operand early termination, per-block clock gating) while the design closed at **zero setup and hold violations across 2,196 paths** and **5.26 mW**.

Cache Replacement Policy & Out-of-Order Study | gem5, C++, Python | 2025

- Implemented the **LRU-IPV** PseudoLRU policy from published research as a C++ SimObject in gem5, keeping strict recency ordering with a shared per-set priority vector at O(associativity) update cost.
- Swept fetch/decode/issue/commit widths and measured a **1.7x** slowdown from 8-wide to 2-wide out-of-order pipelines, identifying memory access as the bottleneck.

Custom CUDA Kernels for Neural Network Training | CUDA, C++, Python | 2025

- Wrote forward and backward CUDA kernels for linear and ReLU layers, integrated them into PyTorch autograd, and exposed a framework-free version to Python through ctypes with explicit host-device memory management.

SVA, Formal & Protocol Verification Program | In Progress, Sept – Dec 2026

- Extending into **SVA** for valid/ready handshakes and FIFOs, a **formal** FIFO safety proof, UVM agents with **RAL** for APB and AXI4-Lite blocks, and a Python/Make regression flow (target: Dec 2026).

EXPERIENCE

Park+ (Parviom Technologies Pvt. Ltd.), Gurugram, India | SDE 2 | Mar 2022 – Jul 2024

- Owned **Linux and Kubernetes infrastructure** for a **60+ microservice** platform across networking, security, and compliance, and built **Shell and Python automation** that cut issue-response time by **20%**.

Arizona State University | Graduate Service Assistant | Aug 2025 – May 2026

- Evaluated EEE 120 digital design labs (adders, multiplexers, decoders, ALU, counters, FSMs) and graded EEE 202 circuit analysis, checking CAD and SPICE simulations for correctness and timing.

Arizona State University | Research Aide, Introduction to Materials Science | May – Aug 2025

- Authored **40+ assignment questions** and technical documentation across 10+ semiconductor topics for a 120+ student cohort.